



Project Title: 🛒 Retail Customer Satisfaction Insights

Duration: 2-3 Hours

🎯 Objective

You are a data analyst tasked with interpreting customer satisfaction trends across retail store locations using NumPy, Pandas, Matplotlib, and Seaborn. Analyze ratings, wait times, and purchasing behavior to deliver key insights.

📁 Dataset

customer_satisfaction_data.csv (generate from AI tool)

📁 Dataset Fields:

- Survey_ID
 - Customer_ID
 - Store_Location
 - Visit_Date
 - Purchase_Amount
 - Wait_Time_Min
 - Staff_Rating
 - Product_Rating
 - Cleanliness_Rating
 - Recommend
 - Total_Rating
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🔴 Tasks Checklist

◆ Data Preparation

1. Load the dataset and display first 5 rows.
 2. Convert Visit_Date into datetime format and extract Month and Weekday.
 3. Replace ratings (1-5) with descriptive categories using NumPy/Pandas (Low, Medium, High).
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◆ Data Analysis

4. Calculate average Total_Rating by Store_Location.
 5. Analyze the correlation between Wait_Time_Min and Purchase_Amount.
 6. Find which day of the week has highest total purchases.
 7. Determine how many customers recommend the store from each location.
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◆ Data Visualization

8. Plot a bar chart of average ratings by store location.
 9. Create a pie chart for recommendation distribution.
 10. Plot a line chart of total purchases over time.
 11. Use Seaborn:
 - Boxplot for Purchase_Amount by Store_Location.
 - Heatmap showing correlation between numeric variables.
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 Submission Requirements:

- Python script or Jupyter notebook with all analysis and visuals.
- Ensure proper labels and clean formatting in charts.